

REVERSAL IS UNREPAIRABLE: EQUIVARIANCE, ELEGANCE, AND THE TEST COMPARISON FOR DE MORGAN CUBICAL SETS

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ABSTRACT. A companion paper identified the quotient of the interval by the reversal as a $K(\mathbb{Z}/2, 1)$ in the cubical-type model structure on De Morgan cubical sets, with contractible geometric realization. This paper closes the two known repair strategies for such defects, and re-opens the question they were meant to settle. First, we determine the automorphism groups of De Morgan and Kleene cubes to be the hyperoctahedral groups and show that the equivariant repair of Awodey, Cavallo, Coquand, Riehl and Sattler carries over: a Quillen model structure with hyperoctahedral-equivariant fibrations exists on De Morgan and Kleene cubical sets, models homotopy type theory, and makes every quotient I^k/H contractible — two modifications are required, since the species-level interval loses all its points and the vertex lemma must be re-proved over orbit groupoids, which in fact simplifies it. But the repair *overshoots*: by the freeness of the reversal action on cells, the category of elements of I^k/H has the homotopy type of BH for every reversal-containing H , so the equivariant model structure contracts objects that the test model structure does not, and — in exact opposition to the cartesian case — cannot present spaces via the test comparison. Second, we prove that the De Morgan and Kleene cube categories admit no relatively elegant embedding into any Reedy category, extending the no-go theorem of Cavallo–Sattler beyond monotone cube maps: the connection-degeneracy freedom that dissolves their winding obstruction at the level of vertices is annihilated by a forcing-and-constancy analysis at the level of free algebras, machine-certified by finite constraint searches; the Boolean cube category, by contrast, stabilizes. Finally, the folklore witness for the non-coincidence of the cubical-type and test model structures on De Morgan cubes evaporates: the test homotopy type of the reversal quotient is $B\mathbb{Z}/2$, in agreement with the cubical-type answer. With both repairs closed and the witness void, whether CCHM cubical type theory presents spaces through the test comparison emerges as a genuinely open problem, for which we record a first positive fragment: the cubical-type weak equivalences form a localizer and hence contain the minimal one.

2020 *Mathematics Subject Classification.* 55U35; 18N40, 55U10, 06D30, 03B38.

Key words and phrases. cubical sets, cubical type theory, model structure, equivariant fibration, test category, Eilenberg–Zilber category, relative elegance, reversal, De Morgan algebra.

1. INTRODUCTION

The cube category $\square_{\text{DM}}$ of CCHM cubical type theory [6] carries connections and a reversal, and its presheaves interpret univalent type theory via the cubical-type model structure [11, 8]. A companion paper [15] proved the folklore of [9, 10]: the quotient $L = \square^1/(x \sim \neg x)$ of the interval by the reversal is not weakly contractible — indeed it is a $K(\mathbb{Z}/2, 1)$ — although its geometric realization is contractible. The present paper asks whether this defect can be *repaired*, by either of the two technologies that have succeeded on neighbouring cube categories:

- *equivariance*: on cartesian cubes, requiring uniform box-filling structures to be equivariant under the automorphisms of the filling cube repairs the analogous defect (the symmetric quotients I^n/H) and yields a model structure that coincides with the test model structure and presents spaces [1];
- *relative elegance*: on cartesian cubes with one connection, a generalized Eilenberg–Zilber theory supplies the cell decompositions on which the comparison with simplicial sets rests [5].

Our answer is that *both* repairs fail for the reversal, each for a structural reason rooted in the freeness theorem of [15]; and that their failure re-opens, rather than settles, the question of the test comparison.

The equivariant route. We determine the symmetry groups first (Theorem 3.1): in the De Morgan and Kleene cube categories, the automorphism group of the n -cube is the hyperoctahedral $G_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$ — for De Morgan by Birkhoff duality and a centralizer computation, for Kleene because the unmixed-poset dual admits no other automorphisms at all. The equivariant machinery of [1] then transfers (Theorem 4.1): a G -equivariant Quillen model structure exists on De Morgan and Kleene cubical sets, models homotopy type theory, and makes every vertex inclusion $1 \rightarrow I^k/H$, $H \leq G_k$, a trivial cofibration — in particular it contracts the reversal quotient L . Two modifications of [1] are needed, and we isolate them precisely: the symmetric-sequence interval loses all its global points under the hyperoctahedral action (the reversal moves every vertex), so the species-level premodel structure is no longer cylindrical and must be demoted to a generation device, with cylindricity, the equivalence extension property, and the fibrant-universe argument running at the cubical-set level — which is possible because the δ -contractor route to fibrant universes is entirely point-free; and the vertex lemma of [1], whose proof there requires the automorphism to *fix* the vertex, is replaced by a colimit over the orbit groupoid, which needs no fixing and is in fact a simplification.

The repair succeeds — and that is exactly its failure. The test model structure on presheaves over a test category has as weak equivalences the maps inverted by the nerve of the category of elements [7, 1]; and by the freeness theorem the reversal acts freely on the *cells* of the interval, so the

category of elements of I^k/H is a free quotient of a contractible category:

$$N\text{el}(I^k/H) \simeq BH \quad (H \leq G_k \text{ acting freely on cells}).$$

Thus the equivariant structure contracts objects whose test homotopy type is $BH \not\simeq *$ (Theorem 5.2): its localizer strictly exceeds the test localizer, the two model structures do *not* coincide — in exact opposition to the cartesian case [1, Thm. 6.3.6] — and the equivariant model cannot present spaces through the test comparison. The comparison with the cartesian case is instructive: there, the symmetric quotients are test-contractible because the *diagonal cells are fixed* by the symmetries, gluing the quotient category into a contractible one; the reversal has no fixed cells — that is the freeness theorem — and the same freeness that makes the covering of [15] a clean fibration makes the equivariant repair overshoot. Equivariance repairs defects born of *fixed* cells; the reversal’s defect is born of their absence.

The elegance route. For the second repair we prove a no-go theorem (Theorem 6.1): *the De Morgan and Kleene cube categories admit no relatively elegant embedding into any Reedy category*. Cavallo–Sattler proved this for Dedekind cubes [5, Thm. A.14] by a winding-number argument over crown posets; their proof uses that Dedekind cube maps are monotone on vertices, and at the vertex level their obstruction genuinely dissolves in $\square_{\text{DM}}$ — De Morgan maps have arbitrary Boolean vertex actions, so the crown factorizations exist as vertex functions. The obstruction resurfaces one level deeper: the required factorization is an equation in free De Morgan algebras, the three-branch structure of the crown extensions forces the factorizing map into the crown fibers, the fibers are pairwise incomparable antichains, and the apparent freedom in the negative-literal directions is annihilated by a constancy argument — monotone maps from a connected poset into an antichain are constant — so the winding obstruction returns. The finite core of the argument is machine-certified: exhaustive constraint searches over the crown regions (768 points for De Morgan, 312 for Kleene) find no admissible factorization, while the Boolean cube category — where vertex actions determine everything — *does* stabilize, consistently with its Karoubi envelope being the Eilenberg–Zilber category of finite sets [4].

The test comparison, re-opened. Finally we revisit the folklore itself. The 2018 discussions record the claim that the cubical-type and test weak equivalences differ on De Morgan cubes [9], with L as the counterexample-in-waiting: its realization is contractible, its cubical-type homotopy type is not. But the test model structure does not see the topological realization; it sees the category of elements, and

$$N\text{el}(L) \simeq B\mathbb{Z}/2 \simeq \text{the cubical-type answer of [15]}.$$

The presumed witness is no witness: on L , the cubical-type and test structures *agree* (and it is the topological realization that is the outlier — it is not a test functor for $\square_{\text{DM}}$). With the witness void and both repair routes closed, we know of no proof that the cubical-type model structure on De

Morgan cubical sets differs from the test model structure; we therefore regard the question

does CCHM cubical type theory present spaces through the test comparison, i.e. is $\mathbf{W}_{\text{type}} = \mathbf{W}_{\text{test}}$ on $\square_{\text{DM}}$?

as open, and record a first fragment (Proposition 7.1): the cubical-type weak equivalences form a Cisinski localizer, hence contain the minimal localizer. The corresponding question for cartesian cubes is settled in the negative by Coquand’s symmetric square [8] — which is consistent, since there the test structure *does* contract the symmetric square (fixed diagonal cells again) while the cubical-type structure does not; the De Morgan situation is genuinely different because the reversal’s freeness aligns the two structures on the known test objects.

Relation to the literature. Theorem 4.1 is an application of the architecture of [1], and we state precisely which components are used unchanged and which are modified; the overshoot phenomenon it runs into is the one [1] themselves flag for BCH cubes in §1.7.2 (“equivariance is not a catch-all solution”), here given an exact combinatorial diagnosis (cell-freeness) and a proof. Theorem 6.1 extends [5, App. A] beyond monotone cube maps. The freeness theorem, the fiber combinatorics of free De Morgan and Kleene algebras, and the identification of L used throughout are from [13, 14, 15]. All finite claims are machine-verified (§8).

2. PRELIMINARIES

We freely use: the cube categories $\square_{\text{DM}}$, $\square_{\text{KL}}$ of the De Morgan and Kleene interval theories, with morphisms $[m] \rightarrow [n]$ the n -tuples of free-algebra elements, and the literal-cube representations of the free algebras [13, 14]; the quotient $L = \square^1 / (t \sim \neg t)$ and the freeness theorem — no cell of the interval is fixed by the reversal — of [15]; the equivariant machinery of [1] (uniform box filling along generalized open boxes $(c : C \multimap I^n, \zeta : I^n \rightarrow I^k)$, with chosen lifts natural in pullback squares *and* in automorphisms σ of the filling cube I^k ; cubical species; Garner’s algebraic small object argument); the test model structure on presheaves over a test category, with weak equivalences the maps f such that $N \text{el}(f)$ is a weak homotopy equivalence [7, 1]; the fact that $\square_{\text{DM}}$, $\square_{\text{KL}}$ and the Boolean cube category are strict test categories [3]; and relative elegance [5]: a Reedy category R is elegant relative to a fully faithful $i : C \rightarrow R$ when the nerve N_i preserves pushouts of lowering spans, which yields cell decompositions of all monomorphisms of $\text{PSh}(C)$ and, via [5, Lem. A.5], a strictly monotone degree on the poset of principal sieves of each object of C .

3. THE AUTOMORPHISM GROUPS ARE HYPEROCTAHEDRAL

Theorem 3.1. *In both the De Morgan and the Kleene cube categories, $\text{Aut}([n]) \cong G_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$, generated by the coordinate permutations and the reversals $x_i \mapsto \neg x_i$.*

Proof. An automorphism of $[n]$ is an automorphism of the free algebra. De Morgan case: by Birkhoff duality, lattice automorphisms of the free De Morgan algebra correspond to order-automorphisms of the literal cube $\{0, 1\}^{2n}$, which are exactly the coordinate permutations Σ_{2n} ; compatibility with the involution $\neg$ (complement composed with the pair swap τ) amounts, since complementation is central, to centralizing τ — a product of n disjoint transpositions, whose centralizer in Σ_{2n} is $\Sigma_n \ltimes (\mathbb{Z}/2)^n$: permute the pairs, swap within a pair, the latter realizing $x_i \mapsto \neg x_i$. Kleene case: the dual object is the unmixed subposet, and a direct check (machine-verified at $n = 2$: all eight poset automorphisms commute with the involution) shows its automorphisms are exactly the same signed permutations; the general statement follows since an automorphism permutes the $2n$ atoms of the poset of principal up-sets respecting pairs, as in the De Morgan case. $\square$

4. THE EQUIVARIANT MODEL STRUCTURE EXISTS

Theorem 4.1. *On De Morgan and on Kleene cubical sets there is a Quillen model structure whose cofibrations are the monomorphisms and whose fibrations are the hyperoctahedral-equivariant uniform fibrations: maps with chosen lifts against the generalized open boxes, uniform in pullback squares and in all $\sigma \in G_k$ acting on the filling cube. It is cylindrical and cartesian closed, satisfies the Frobenius condition, the equivalence and fibration extension properties, and has univalent universes with fibrant bases — a model of homotopy type theory. Moreover, for every $k \geq 1$, every subgroup $H \leq G_k$, and every vertex v of I^k , the inclusion $v : 1 \rightarrow I^k/H$ is a trivial cofibration; in particular $L \rightarrow 1$ is a weak equivalence.*

Proof (outline, with the two modifications). The construction follows §§2–5 of [1], with the groupoid Σ replaced by the hyperoctahedral groupoid $G = \coprod_k G_k$ (legitimate by Theorem 3.1): G -species, the symmetric-sequence interval $\mathbb{I}$ (tiny, since the cube categories have finite decidable homs — free De Morgan and Kleene algebras on finitely many generators are finite), the generating category of boxes, Garner’s algebraic small object argument, and the orbit transfer along $L \dashv \Delta$. Two points do not transfer verbatim.

First modification (pointless species interval). A global point of $\mathbb{I}$ chooses a G_k -fixed vertex of I^k for each k ; under the hyperoctahedral action there are none (the reversal moves every vertex — the freeness theorem at the vertex level), so the species-level premodel structure is not cylindrical, and the species-level model structure of [1, §4.4] is unavailable. It is also unneeded: the cubical-set-level architecture of [1, §5] uses the species layer only as a generation device. Cylindricity at the cubical-set level needs the endpoint inclusions $0, 1 : 1 \rightarrow I$ to be trivial cofibrations, which are the $k = 1$ vertex-triangle generators and require no equivariance; the pushout-product axiom transfers along Δ (which preserves Leibniz exponentials); the equivalence extension property is proved at the cubical-set level from cylindricity (axiomatic); Frobenius transfers along the pushforward-preserving Δ ; and the

fibrant-universe argument runs through the δ -contractor mechanism of [1, §3.6], which is entirely point-free — it uses only the generic point $\delta : \mathbb{I} \rightarrow \mathbb{I} \times \mathbb{I}$ in the slice, never an endpoint of $\mathbb{I}$. The fibration extension property and two-out-of-three then follow as in [1, §5.3].

Second modification (orbit-groupoid vertex lemma). In [1, Lem. 5.2.5] the contractibility of I^k/H rests on morphisms of vertex triangles $(\text{id}, \sigma) : (\emptyset \rightharpoonup 1, \sigma^{-1}\vec{v}) \rightarrow (\emptyset \rightharpoonup 1, \vec{v})$, and their diagram is constant at one triangle, which requires $\sigma^{-1}\vec{v} = \vec{v}$ — only the two poles are fixed by all permutations. We instead organize the H -action as a diagram over its *orbit groupoid*: objects the vertex triangles at the vertices of one orbit, morphisms the (id, σ) , with no fixing condition (the constraint $\xi = \sigma^{-1}\zeta\alpha$ of the generating category is then the definition of the diagram, not a restriction). The colimit of this diagram of generating trivial cofibrations has domain $\text{colim}(1) = 1$ (connected groupoid) and codomain I^k/H , and coalgebra forgetful functors create colimits [2], so $v : 1 \rightarrow I^k/H$ is a trivial cofibration for *every* vertex — a simplification as well as a generalization of the original lemma. For $L = I/\langle \neg \rangle$ this gives $1 \rightarrow L$ a trivial cofibration, whence $L \rightarrow 1 \in \mathcal{W}_{\text{eq}}$ by two-out-of-three. $\square$

5. THE REPAIR OVERSHOTS

Lemma 5.1. *Let $H \leq G_k$ act freely on the cells of I^k (by the freeness theorem, any H containing a reversal-component acts freely already through that component). Then $N\text{el}(I^k/H) \simeq BH$; in particular $N\text{el}(L) \simeq B\mathbb{Z}/2$.*

Proof. $\text{el}(-)$ preserves colimits, so $\text{el}(I^k/H) = \text{el}(I^k)/H$, a quotient of a category with terminal object (the slice over a representable) by an action free on objects. Freeness on objects makes the action free on all composable strings, so the quotient functor is a covering of categories and $N(\text{el}(I^k)/H) = N(\text{el}(I^k))/H$, the quotient of a contractible free H -complex: a BH . $\square$

Theorem 5.2. *On De Morgan and Kleene cubical sets, the hyperoctahedral-equivariant model structure does not coincide with the test model structure: $L \rightarrow 1 \in \mathcal{W}_{\text{eq}} \setminus \mathcal{W}_{\text{test}}$. Consequently the equivariant model structure does not present spaces through the test comparison — in exact opposition to the cartesian case [1, Thm. 6.3.6].*

Proof. $L \rightarrow 1 \in \mathcal{W}_{\text{eq}}$ by Theorem 4.1; $N\text{el}(L) \simeq B\mathbb{Z}/2 \not\simeq *$ by Lemma 5.1, so $L \rightarrow 1 \notin \mathcal{W}_{\text{test}}$. Since $\square_{\text{DM}}$ is a strict test category [3], the test model structure presents spaces [7]; a model structure with the same cofibrations and a strictly larger class of weak equivalences on a contractible-realization object is a different localization. $\square$

Remark 5.3. The diagnosis, in one line: *equivariance repairs defects born of fixed cells, and the reversal has none.* On cartesian cubes the symmetric quotients I^n/H are test-contractible — their element categories are glued along the fixed diagonal cells — so enlarging the localizer to contract them moves the cubical-type structure *toward* the test structure, and [1] shows it

lands exactly on it. On De Morgan cubes the freeness theorem makes the reversal quotients test-nontrivial (BH 's), so the same enlargement moves *past* the test structure. This realizes, with a proof, the caution of [1, §1.7.2] that “equivariance is not a catch-all solution”: the BCH cube category was their suspected example; the De Morgan cube category is a proven one.

6. NO RELATIVELY ELEGANT EMBEDDING

We now close the second route. Recall from [5, App. A] the crown posets Cr_m (cyclic fences of $2m$ elements), the embeddings $c_m : \text{Cr}_m \rightarrow [1]^m$, the wrapping maps $f_a : \text{Cr}_{am} \rightarrow \text{Cr}_m$ of winding number a , and the extensions $\bar{f} : [1]^{am} \rightarrow [1]^m$ sending $c_{am}(i) \mapsto c_m(f(i))$, $\perp \mapsto \perp$, and everything else to $\top$; the three branches have disjoint values, so the defining square is a pullback. If some $i : \mathcal{C} \rightarrow R$ were relatively elegant, principal sieves on each object of $\mathcal{C}$ would carry a strictly monotone degree [5, Lem. A.5] — the lemma’s fullness hypothesis is part of the definition of relative elegance, so it applies to any $\mathcal{C}$ — and the chain $\langle \bar{f}_2 \rangle \supseteq \langle \bar{f}_4 \rangle \supseteq \cdots$ of principal sieves on $[1]^n$ would stabilize: some $\bar{f}_a = \bar{f}_b \circ g$ with $a < b$ and $g : [1]^{an} \rightarrow [1]^{bn}$ a map of $\mathcal{C}$.

Theorem 6.1. *For $\mathcal{C} \in \{\square_{\text{DM}}, \square_{\text{KL}}\}$, no such g exists; hence the De Morgan and Kleene cube categories admit no relatively elegant embedding into any Reedy category.*

Proof. A first phenomenon, absent from the Dedekind case: at the level of *vertex actions* the factorization exists. De Morgan cube maps have arbitrary Boolean vertex actions (disjunctive normal forms over literals are functionally complete), the three branches of $\bar{f}_b$ are disjoint, and a set-level section of f_b over the crown extended by $\perp \mapsto \perp$, else $\top$, satisfies the vertex-level equation — no monotonicity, hence no winding number, obstructs it. The obstruction returns at the free-algebra level, where the required equation $\bar{f}_b \circ g = \bar{f}_a$ holds in free algebras, i.e. as monotone functions on the literal cube. Write G for the induced monotone map on literal points and pos for the projection to positive literals. Since $\bar{f}_b$ is negation-free, the composite’s literal function is $\bar{f}_b$ ’s vertex action applied to G ; since $\bar{f}_b^{-1}(\text{crown values}) = \text{crown points}$ exactly, G is forced, on the region $\{\alpha : \text{pos}(\alpha) \in c_{an}(\text{Cr}_{an})\}$, to take values in c_{bn} of the fibers $f_b^{-1}(f_a(i))$ — antichains of b points spaced $2n$ apart in Cr_{bn} , pairwise incomparable. The region is the product of the crown subposet with the Boolean cube of negative literals; the negative directions are connected, and a monotone map from a connected poset into an antichain is constant, so G ’s crown-region behaviour factors through a monotone $\rho : \text{Cr}_{an} \rightarrow \text{Cr}_{bn}$ with $f_b \rho = f_a$ — and $b \cdot \text{wind}(\rho) = a$ with $a < b$ has no integer solution. (Machine certificates: the crown-region constraint problems — 768 points for De Morgan, 312 for Kleene, both with domains of size $b = 4$ — are exhaustively unsatisfiable; §8.) For Kleene the same argument runs on the unmixed literal points, whose crown region remains connected in the negative directions. $\square$

Remark 6.2. The Boolean cube category escapes: there the free algebras are determined by vertex actions, the set-level section *is* a cube map, the chain stabilizes immediately, and indeed the Boolean Karoubi envelope is the Eilenberg–Zilber category of finite nonempty sets [4]. The trichotomy Boolean/Kleene–De Morgan reproduces, at the level of elegance, the fiber trichotomy of [14]: what obstructs is precisely the monotone literal structure that the involution fails to collapse.

7. THE TEST COMPARISON, RE-OPENED

Proposition 7.1. *On any of the three cube categories, the weak equivalences W_{type} of the cubical-type model structure form a Cisinski localizer: they satisfy two-out-of-three, contain the maps with the right lifting property against all monomorphisms, and their monic members are closed under pushout and transfinite composition. Consequently $W_{\text{min}} \subseteq W_{\text{type}}$, where W_{min} is the minimal localizer.*

Proof. All three properties are theorems of model-category theory for a model structure whose cofibrations are the monomorphisms; the minimal localizer is contained in every localizer. $\square$

Remark 7.2 (the folklore witness is void). The 2018 record asserts that the cubical-type and test weak equivalences differ on De Morgan (and Boolean) cubes [9], with the reversal quotient as the underlying example. But L separates the cubical-type structure from the *topological realization*, not from the test structure: by Lemma 5.1 its test homotopy type is $B\mathbb{Z}/2$, which is precisely its cubical-type homotopy type [15]. (It follows, incidentally, that the topological cocube on $\square_{\text{DM}}$ is not a test functor: it computes $*$ where the test structure computes $B\mathbb{Z}/2$.) On the cartesian side the analogous question is genuinely settled in the negative — Coquand’s symmetric square is cubical-type-nontrivial [8] yet test-trivial (the equivariant structure contracts it and coincides with the test structure [1]) — but that argument turns on the fixed diagonal cells, which the reversal lacks. We know of no surviving proof that $W_{\text{type}} \neq W_{\text{test}}$ on $\square_{\text{DM}}$, and we regard the question — *does CCHM cubical type theory present spaces through the test comparison?* — as open. Proposition 7.1 gives the lower bound; the missing steps for the upper bound $W_{\text{type}} \subseteq W_{\text{test}}$ are the test-triviality of the generalized open boxes (plausible: their element categories are glued from contractible pieces along fully faithful inclusions) and, for equality, a converse that would necessarily be as hard as the known open Dedekind case [12, 5].

8. MACHINE VERIFICATION

Finite claims verified exhaustively (scripts accompany the artifact): the automorphism counts of Theorem 3.1 ($8 = |G_2|$ coordinate permutations commuting with the involution for De Morgan at $n = 2$; all 8 poset automorphisms of the unmixed poset, all commuting, for Kleene); the freeness

census underlying Lemma 5.1 (0 self-dual cells through three generators in both theories, from [15]); the non-commutation $(\neg x) \wedge e \neq \neg(x \wedge e)$ showing hyperoctahedral equivariance is not implied by connections (the connection-contraction argument for automatic symmetric-group equivariance [5, Rem. 4.25] commutes with permutations only); and the crown-region constraint problems of Theorem 6.1: for $(n, a, b) = (3, 2, 4)$, the De Morgan region has 768 points with 16,728 order constraints and the Kleene region 312 points, all domains of size 4; backtracking with propagation finds no solution in either, while the fiber-incomparability and three-branch-disjointness lemmas are checked directly.

9. OUTLOOK

The results funnel further work toward one question: the upper bound $W_{\text{type}} \subseteq W_{\text{test}}$ of Remark 7.2, whose proof would need the test-triviality of the generalized open boxes of $\square_{\text{DM}}$, and whose converse — equality — would settle whether CCHM cubical type theory presents spaces. Both halves are now unobstructed by any known counterexample. Independently, the equivariant model structure of Theorem 4.1 presents *some* homotopy theory — a localization of the test theory in which the classifying spaces BH of reversal-containing subgroups are contracted — whose identification we leave open; and the orbit-groupoid form of the vertex lemma may be of use wherever the ACCRS machinery meets non-vertex-fixing symmetries. Artifact. The verification scripts and the machine-checked companions are available in the artifact repository:

<https://github.com/r-shibusawa/cubical-strict-j>, archived at DOI 10.5281/zenodo.21405961.

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